

Matrices

↳ Pure Basics :

Row Matrix $\rightarrow [1 \ 2 \ 3]_{1 \times 3} \ (m=1)$

$$\begin{array}{cc} & m \times n \\ \swarrow & \downarrow \\ \text{Row} & \text{Column} \end{array}$$

Column Matrix $\rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3 \times 1} \ (n=1)$

Rectangular Matrix $\rightarrow \begin{bmatrix} 1 & 2 & 4 & 5 \\ 2 & 0 & 3 & 8 \\ 7 & 4 & 2 & 5 \end{bmatrix}_{3 \times 4} \ (m \neq n)$

Square Matrix $\rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}_{3 \times 3}$

Diagonal Matrix $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$

↳ $m=n$

↳ All elements except the diagonal corner are zero

Scalar Matrix $\rightarrow \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}_{3 \times 3}$

↳ $m=n$

↳ All elements of diagonal corner must be equal and the rest are zero

Identity Matrix $\rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

↳ $m=n$

↳ All elements of diagonal corner must be one and the rest are zeros.

Combination of column $\rightarrow A_n = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}_{3 \times 2} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2 \times 1} \Rightarrow Ax = x_1 \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}_{3 \times 1} + x_2 \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}_{3 \times 1}$

Solution of $Ax=b$ can be written as $x = A^{-1}b$

Writing System (From Kimberly)

Ex 1: $\begin{cases} x_1 + x_2 = 5 \\ 3x_1 - 2x_2 = 10 \end{cases}$ can be expressed as $\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix}$, also $\begin{bmatrix} 1 & 1 & 5 \\ 3 & -2 & 10 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 \\ 3 & -2 \end{bmatrix}$
co-efficient matrix

$\begin{bmatrix} 1 & 1 & 5 \\ 3 & -2 & 10 \end{bmatrix}$
augmented matrix

$$\begin{bmatrix} 1 & 1 & 5 \\ 3 & -2 & 10 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 2 & 10 \\ 3 & -2 & 10 \end{bmatrix} \quad R_2 \rightarrow 2R_1$$

$$\Rightarrow \begin{bmatrix} 5 & 0 & 20 \\ 3 & -2 & 10 \end{bmatrix} \quad R_1 \rightarrow R_2$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 4 \\ 3 & -2 & 10 \end{bmatrix} \quad \frac{1}{5} R_1$$

$$\Rightarrow \begin{bmatrix} -3 & 0 & -12 \\ 3 & -2 & 10 \end{bmatrix} \quad -3R_1$$

$$\begin{aligned}
 &= \begin{bmatrix} -3 & 0 & 1 & -12 \\ 3 & -2 & 1 & 10 \end{bmatrix} \quad -3R_1 \\
 &= \begin{bmatrix} -3 & 0 & 1 & -12 \\ 0 & -2 & 1 & -2 \end{bmatrix} \quad R_1 + R_2 \\
 &= \begin{bmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 1 \end{bmatrix} \quad \begin{aligned} &\hookrightarrow x_1 = 4 \\ &x_2 = 1 \end{aligned}
 \end{aligned}$$

Ex 2:

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -4x_1 + x_2 + x_3 = -2 \end{cases} \quad \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 1 & 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 1 & 1 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 5 & -2 \end{bmatrix} \quad R_3 - 4R_1 + R_2$$

$$= \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 5 & -2 \end{bmatrix} \quad R_2 = \frac{1}{2}R_2$$

$$= \begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{bmatrix} \quad R_3 = 3R_2 + R_3$$

$$\hookrightarrow \begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ x_3 = 3 \end{cases} \quad \begin{aligned} &x_1 = 29 \\ &x_2 = 16 \\ &x_3 = 3 \end{aligned}$$

⊛ A solution will be inconsistent if there's a form of $0x_1 + 0x_2 + 0x_3 = 0$.
Thus no solution exists.

Echelon Form

↳ loosely defined term referring to any form of a matrix where the leading co-efficients (non-zero entries) in each row are staggered to the right.

$$\begin{bmatrix} 1 & 3 & 4 \\ 0 & 2 & 5 \\ 0 & 0 & 6 \end{bmatrix}$$

Row Echelon Form (REF)

↳ Any row consisting entirely of zeros occurs at the bottom of the matrix.

↳ Each leading entry of a row is in a column to the right of the leading entry of the row above it. The leading value is 1.

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

... column below a leading entry are zeros

leading entry of the row above it, the remaining ones are 1.

↳ All entries in a column below a leading entry are zeros

Reduced Row Echelon Form (RREF)

↳ Each leading 1 is the only non-zero entry in the column.

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

⊗ Any matrix can be transformed to RREF using Gaussian Elimination Method.

Pivot

↳ First non-zero element in a row.

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 1 & 1 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

↗ pivots

⊗ Will be continued after Gaussian Elimination.

Matrices as linear transformation

↳ can be
think as a
function

Properties of linear transformation

↳ All lines must remain lines without getting curved.

↳ The origin must remain fixed

In general, the grid lines are parallel and evenly spaced.

Example: $i \rightarrow \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad j \rightarrow \begin{bmatrix} 3 \\ 0 \end{bmatrix}$

$$\hookrightarrow \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow x \begin{bmatrix} -1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -x + 3y \\ 2x + 0y \end{bmatrix}$$

$$\hookrightarrow \begin{bmatrix} 1 \\ -2 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} \rightarrow \text{Where } \hat{j} \text{ lands}$$

$\rightarrow$ Where $\hat{i}$ lands

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\hookrightarrow x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}$$

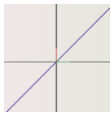
$$= \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}$$

Visual Examples:

Ex 1:

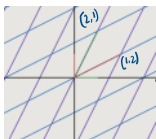
$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$


Ex 2:

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow$$


$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \nearrow$$

Ex 3:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$


$\rightarrow$ evenly spaced parallel lines

⊗ If there are only one line after linear transformation, then it is linearly dependent.

Linear Dependence & Independence

$\hookrightarrow$ In details:

Linear Independence

A set of vectors is said to be **linearly independent** if none of the vectors in the set can be written as a

linear combination of the others. In other words, the only solution to the equation:

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$$

is when all the scalar coefficients $c_1, c_2, \dots, c_n$ are **zero**. This means that the vectors are completely independent of each other, and no vector in the set depends on another.

Geometrically:

- In 2D space, two vectors are linearly independent if they do not lie on the same line.
- In 3D space, three vectors are linearly independent if they do not all lie on the same plane.

2. Linear Dependence

A set of vectors is **linearly dependent** if at least one of the vectors can be written as a linear combination of the others. In this case, there exists a **non-trivial** solution to the equation:

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$$

where **not** scalar coefficients $c_1, c_2, \dots, c_n$ are **zero**. This means that the vectors are **dependent** on each other, and at least one vector is "redundant" in the sense that it can be expressed in terms of the others.

Geometrically:

- In 2D space, two vectors are linearly dependent if they lie on the same line.
- In 3D space, three vectors are linearly dependent if they lie in the same plane.

Column Space

↳ The range of the linear transformation described by the matrix

if $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$

↳ so the column space would go like:

$$A = c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + c_3 \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

Each of these columns can be represented as a vector.

Rank of the Matrix

↳ The dimension of the column space is called the rank of the matrix.

It is equal to the number of linearly independent columns in the matrix.

Rank 1 Matrix

↳ All columns are multiples of one column, and similarly, all rows are multiples of one row.

So row rank = column rank

Ex: $A = \begin{bmatrix} 1 & 3 & -2 \\ 4 & 12 & -8 \\ 2 & 6 & -4 \end{bmatrix}$

Problem Set 1.3

1. $A = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

$$1. A_1 = \begin{bmatrix} 2 & 2 \\ 1 & 1 \\ 5 & 6 \end{bmatrix} = C_1 \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix} + C_2 \begin{bmatrix} 2 \\ 1 \\ 6 \end{bmatrix}$$

↳ Not scalar multiple of each other. So, they are linearly independent.
That give us a plane.

$$A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \rightarrow \text{three independent column space}$$

↳ space all of 3D space.

$$A_3 = \begin{bmatrix} 1 & 5 \\ 2 & 10 \\ 1 & 5 \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} + C_2 \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix} = C_1 u_1 + C_2 u_2$$

Here, $u_2 = 5u_1$, so they are linearly dependent. That gives us one independent column space. That give us a line in 3D space.

$$A_4 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{corresponds to a point in space.}$$

$$2. A_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow \cancel{u} \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix} + \cancel{v} \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix} + \cancel{w} \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{array} \right] \quad C_2 \leftarrow 2C_1$$

$$= \left[\begin{array}{ccc|c} 1 & 3 & 3 & 0 \\ 4 & 6 & 6 & 0 \\ 7 & 9 & 9 & 0 \end{array} \right] \quad C_2 \leftarrow C_2 - C_1$$

$$= \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 4 & 6 & 0 & 0 \\ 7 & 9 & 0 & 0 \end{array} \right] \quad C_3 \leftarrow C_3 - C_2$$

↳ inconsistent.

$$\begin{matrix} 0 & 0 \\ 0 & 0 \end{matrix} \quad 2C_2 - C_1 = C_3 \rightarrow \text{This combination produces } (0,0,0)$$

$\therefore$ 1 column are dependant on first and second column.

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0 0 ~ 2 . /

Third column are dependant on first and second column.